

Assignment: IPL Data Platform (Full-Stack – Intern Profile)

Overview

You are given an IPL (Indian Premier League) dataset and asked to design and build a production-ready full-stack application. This assignment evaluates your ability to work across data modeling, backend and frontend development, containerization, CI/CD, and cloud-native deployment practices.

The application should:

1. Store the data in PostgreSQL
2. Expose data via Backend APIs (Node.js or Python)
3. Document APIs using OpenAPI (Swagger)
4. Display insights on a Frontend web application
5. Be containerized and deployed using modern DevOps practices (Far fetched goal)

Dataset

The IPL dataset can be downloaded from the link below:

[Indian Premier League 2022-03-26.zip](#)

You are expected to understand the dataset, design an appropriate schema, and manage data ingestion in a scalable manner.

Tech Stack (Flexible)

Database: PostgreSQL

Backend: Node.js or Python

Data Access: ORM or query builder of your choice (Prisma, Drizzle, SQLAlchemy, etc.)

Frontend: React / Next.js or equivalent

Containers: Docker

CI/CD: GitHub Actions

Cloud: Any major cloud provider (Azure or GCP preferred)

Core Requirements

1. Database & Data Modeling

Design a relational schema suitable for the IPL dataset. Use migrations and seed/load the provided data into PostgreSQL.

2. Backend API

Build backend APIs that expose IPL data to the frontend. APIs must return JSON, support pagination and filtering where applicable, include proper validation and error handling, and expose a health check endpoint. All APIs must be documented using OpenAPI, with Swagger UI enabled.

3. Frontend Application

Create a web UI that consumes backend APIs and presents IPL insights. The application must include multiple screens/pages, charts and tabular views, and must properly handle loading, empty, and error states.

4. Containerization

Create Dockerfiles for backend and frontend services. Provide a docker-compose setup for local development.

5. CI/CD

Set up GitHub Actions workflows to run linting and tests, build Docker images, and deploy applications on successful merges.

Optional Stretch Goals

- Terraform code for infrastructure provisioning (Infrastructure as Code)
- Datadog integration for monitoring, logging, and basic dashboards
- Kubernetes deployment, with proper configmaps and helm charts

Submission Checklist

GitHub repository containing:

- Clear README with architecture overview
- Local and deployed setup instructions
- Database schema/migrations
- Dockerfiles and docker-compose
- GitHub Actions workflows
- Kubernetes configuration files
- Access to OpenAPI documentation
- Deployed application URLs

Notes

This assignment is intended for experienced engineers. Code quality, system design, clarity of thought, and production readiness will be valued more than feature count.